

Task 1: Data Modeling & Cleaning

- Load and transform datasets in Power Query

	A _C Transaction_ID	A _C Customer_ID	A _C Store_ID	A _C Product_Category	Transaction_Date	L2 Amount	A _C Promotion_Applied	i ₃ Year	A _C Month Name
1	T0531	C001	S20	Apparel	19-10-2024	335.04	Yes	2024	October
2	T0597	C001	S09	Groceries	11-06-2024	551.76	Yes	2024	June
3	T0454	C001	S12	Home & Living	16-03-2023	527.9	No	2023	March
4	T0857	C001	S16	Apparel	20-06-2024	545.13	No	2024	June
5	T0550	C001	S02	Electronics	02-07-2024	765.62	Yes	2024	July
6	T0818	C001	S14	Electronics	23-10-2024	632.38	No	2024	October
7	T0274	C001	S19	Home & Living	08-12-2023	461.93	Yes	2023	December
8	T0356	C002	S18	Electronics	24-03-2024	236.76	Yes	2024	March
9	T0836	C002	S11	Groceries	18-02-2024	362.58	No	2024	February
10	T0580	C002	S02	Apparel	27-05-2023	262.28	Yes	2023	May
11	T0800	C002	S07	Apparel	14-03-2023	361.94	No	2023	March
12	T0024	C002	S14	Groceries	25-03-2024	103.15	Yes	2024	March
13	T0805	C003	S17	Groceries	26-10-2023	213.1	Yes	2023	October
14	T0797	C003	S01	Apparel	05-01-2024	599.4	Yes	2024	January
15	T0776	C004	S06	Groceries	23-07-2024	108.38	No	2024	July
16	T0506	C004	S20	Apparel	21-08-2023	329.67	Yes	2023	August
17	T0530	C005	S15	Home & Living	21-10-2023	679.57	No	2023	October
18	T0344	C005	S16	Groceries	29-05-2023	608.93	No	2023	May
19	T0355	C005	S13	Apparel	06-08-2024	212.41	No	2024	August
20	T0023	C005	S09	Groceries	31-12-2023	140.64	No	2023	December
21	T0758	C005	S09	Electronics	14-08-2024	122.74	No	2024	August
22	T0798	C006	S02	Home & Living	21-05-2024	85.76	No	2024	May
23	T0025	C006	S07	Apparel	21-06-2024	50.75	No	2024	June
24	T0651	C006	S03	Apparel	16-05-2024	33.22	No	2024	May
25	T0243	C006	S01	Electronics	06-06-2024	200.92	Yes	2024	June
26	T0631	C006	S15	Home & Living	06-07-2024	391.25	Yes	2024	July
27	T0202	C006	S01	Apparel	07-08-2024	90.58	Yes	2024	August
28	T0710	C007	S09	Electronics	02-08-2024	473.67	No	2024	August
29	T0210	C007	S05	Groceries	11-05-2024	366.65	Yes	2024	May
30	T0446	C007	S07	Groceries	20-04-2024	689.38	No	2024	April
31	T0883	C007	S10	Apparel	08-08-2023	324.79	Yes	2023	August
32	T0122	C007	S14	Groceries	11-07-2024	380.44	Yes	2024	July
33	T0669	C007	S10	Apparel	16-02-2024	521.4	No	2024	February
34	T0267	C007	S03	Home & Living	31-03-2024	609.11	Yes	2024	March
35	T0743	C008	S04	Apparel	27-02-2024	380.14	No	2024	February
36	T0934	C008	S07	Apparel	12-02-2023	373.79	No	2023	February
37	T0499	C008	S30	Apparel	09-10-2024	206.31	No	2024	October
38	T0078	C009	S10	Groceries	04-05-2023	310.7	Yes	2023	May

- Handle duplicates, missing values, and ensure correct data types

	A _C Customer_ID	i ₃ Age	A _C Gender	A _C Region	A _C Income_Level	Membership_Since	A _C Preferred_Channel	i ₃ Membership_Duration
1	C001		58 Female	North	Low	18-08-2023	Online	853
2	C002		48 Male	North	High	17-09-2019	Store	2284
3	C003		34 Female	East	Medium	06-03-2023	Store	1018
4	C004		27 Female	Central	Low	30-06-2021	Store	1632
5	C005		40 Male	South	High	12-09-2022	Online	1193
6	C006		58 Male	South	High	30-10-2018	Online	2606
7	C007		38 Female	East	High	01-10-2019	Store	2270
8	C008		42 Female	South	Low	25-02-2018	Online	2853
9	C009		30 Male	North	Medium	27-12-2019	Online	2183
10	C010		30 Female	Central	Medium	10-06-2019	Online	2383
11	C011		43 Male	West	Medium	19-04-2019	Store	2435
12	C012		55 Female	South	High	28-01-2018	Store	2881
13	C013		59 Male	North	Medium	27-10-2023	Online	783
14	C014		43 Male	West	High	31-12-2018	Store	2544
15	C015		22 Male	Central	High	10-06-2019	Online	2383
16	C016		41 Female	West	High	28-08-2021	Online	1573
17	C017		21 Female	North	Medium	01-06-2020	Online	2026
18	C018		43 Male	West	Low	27-10-2020	Online	1878
19	C019		49 Female	East	Medium	23-10-2020	Store	1882
20	C020		57 Male	West	High	05-07-2022	Store	1262
21	C021		21 Male	South	Low	07-07-2020	Store	1990
22	C022		40 Female	South	Medium	07-08-2024	Online	498
23	C023		52 Female	East	Low	10-11-2022	Online	1134
24	C024		31 Male	North	Medium	12-03-2022	Online	1377
25	C025		41 Female	South	High	22-03-2023	Store	1002
26	C026		44 Female	Central	High	30-04-2021	Online	1693
27	C027		46 Male	South	Low	13-07-2022	Store	1254
28	C028		47 Male	South	High	30-04-2019	Store	2424

- Create calculated columns:
 - o Membership_Duration = Today – Membership_Since

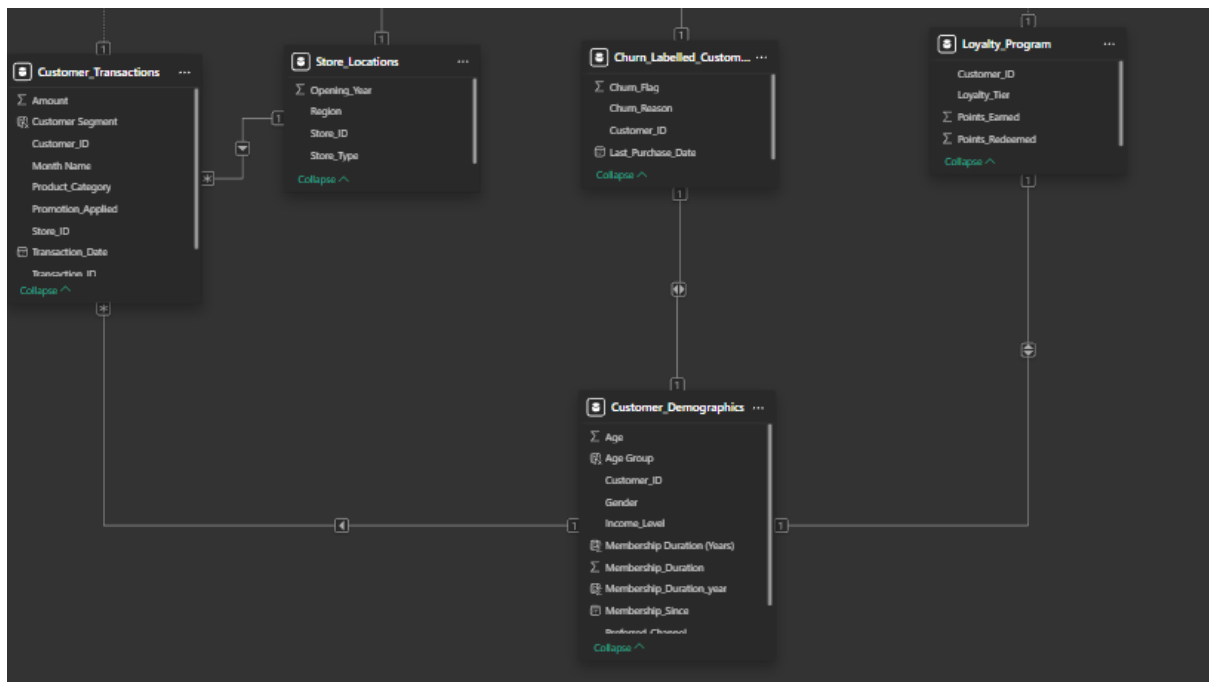
Membership_Since	Preferred_Channel	Membership_Duration
18-08-2023	Online	853
17-09-2019	Store	2284
06-03-2023	Store	1018
30-06-2021	Store	1632
12-09-2022	Online	1193
30-10-2018	Online	2606
01-10-2019	Store	2270
25-02-2018	Online	2853
27-12-2019	Online	2183
10-06-2019	Online	2383
19-04-2019	Store	2435
28-01-2018	Store	2881
27-10-2023	Online	783
31-12-2018	Store	2544
10-06-2019	Online	2383
28-08-2021	Online	1573
01-06-2020	Online	2026
27-10-2020	Online	1878
23-10-2020	Store	1882
05-07-2022	Store	1262
07-07-2020	Store	1990
07-08-2024	Online	498
10-11-2022	Online	1134
12-03-2022	Online	1377
22-03-2023	Store	1002
30-04-2021	Online	1693
13-07-2022	Store	1254
30-04-2019	Store	2424
07-09-2023	Store	833
15-12-2019	Store	2195
28-11-2020	Store	1846
31-05-2023	Store	932
20-07-2021	Online	1612
08-06-2021	Online	1654
04-11-2023	Store	775
06-05-2018	Store	2783
21-06-2019	Online	2372
02-01-2024	Store	716

- o Extract Transaction_Year, Transaction_Month

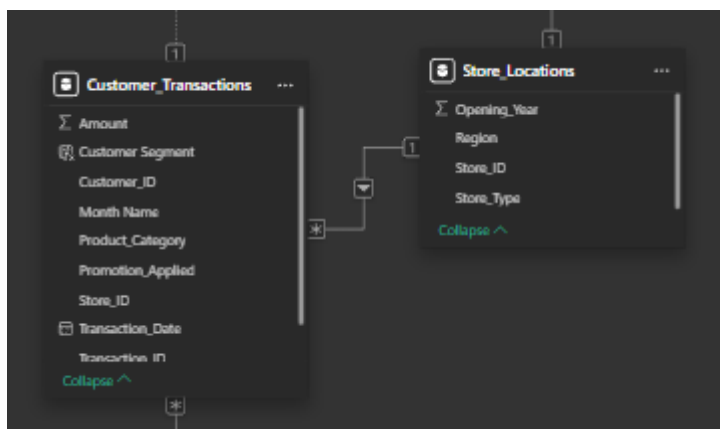
Product_Category	Transaction_Date	Year	Month Name
Apparel	20-06-2024	2024	June
Home & Living	16-03-2023	2023	March
Electronics	23-10-2024	2024	October
Electronics	02-07-2024	2024	July
Apparel	19-10-2024	2024	October
Groceries	11-06-2024	2024	June
Home & Living	08-12-2023	2023	December
Electronics	24-03-2024	2024	March
Groceries	18-02-2024	2024	February
Apparel	27-05-2023	2023	May
Apparel	14-03-2023	2023	March
Groceries	25-03-2024	2024	March
Groceries	26-10-2023	2023	October
Apparel	05-01-2024	2024	January
Apparel	21-08-2023	2023	August
Groceries	23-07-2024	2024	July
Home & Living	21-10-2023	2023	October
Electronics	14-08-2024	2024	August
Apparel	06-08-2024	2024	August
Groceries	29-05-2023	2023	May
Groceries	31-12-2023	2023	December
Home & Living	06-07-2024	2024	July
Home & Living	21-05-2024	2024	May
Electronics	06-06-2024	2024	June
Apparel	07-08-2024	2024	August
Apparel	16-05-2024	2024	May
Apparel	21-06-2024	2024	June
Apparel	08-08-2023	2023	August
Groceries	11-05-2024	2024	May
Apparel	16-02-2024	2024	February
Home & Living	31-03-2024	2024	March
Groceries	11-07-2024	2024	July
Groceries	20-04-2024	2024	April
Electronics	02-08-2024	2024	August
Apparel	27-02-2024	2024	February

- Create a basic Data Model view

- One-to-Many: Customer_Demographics → Transactions, Loyalty_Program, Churn_Labelled_Customers



- Many-to-One: Transactions → Store_Locations



Task 2: Churn & Retention Metrics

- Create Churn Rate KPI = (Churned Customers / Total Customers) * 100

24.50
Churn Rate (%)

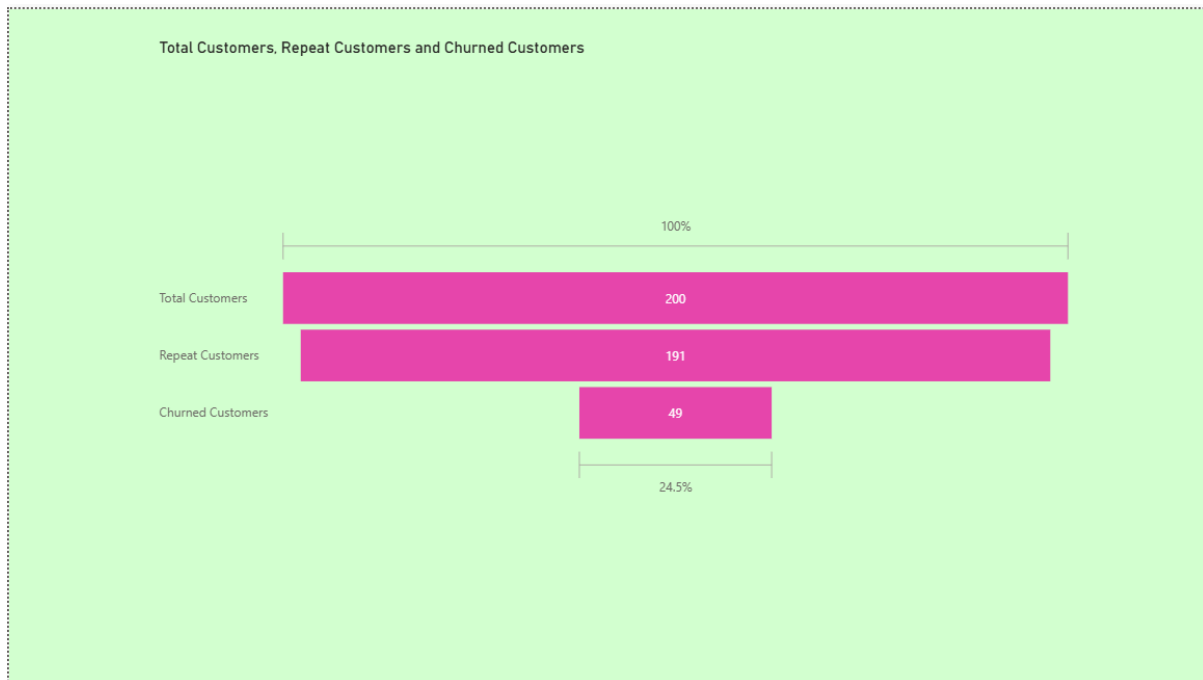
Churn Rate (%) =

```
DIVIDE(  
  [Churned Customers],  
  [Total Customers],  
  0  
) * 100
```

- Visualize churn rate by:



- Funnel Chart: Total Customers → Repeat Customers → Churned



Task 3: Repeat Purchase Analysis

- Segment customers:

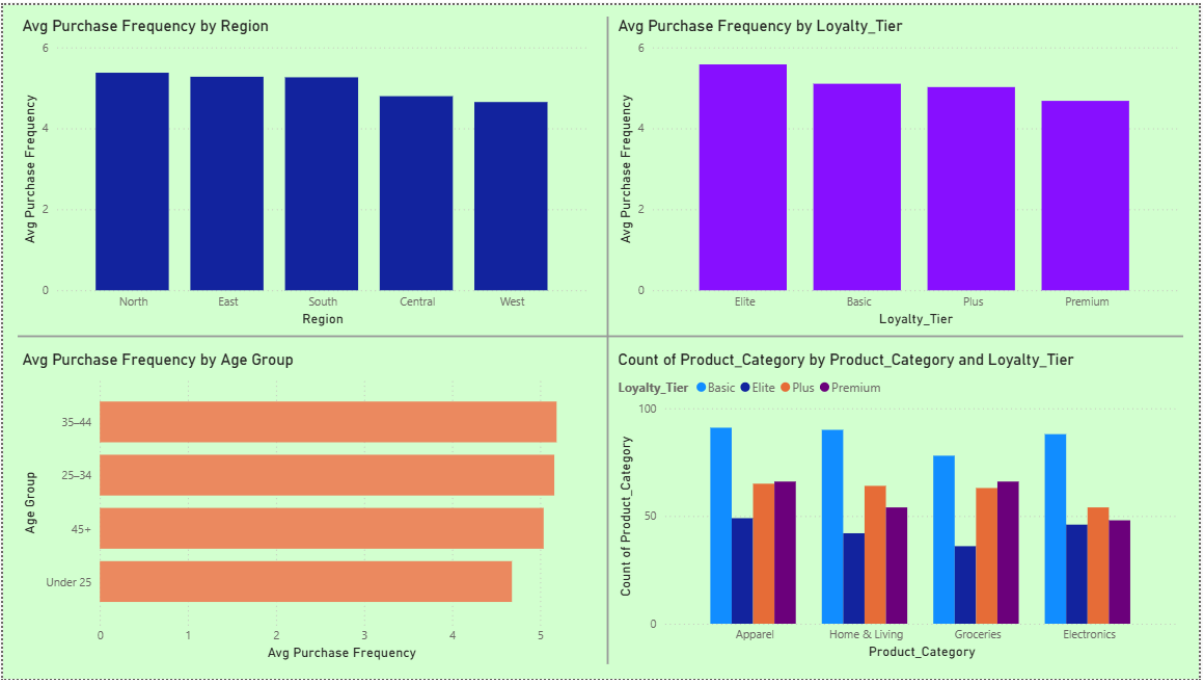
Customer_ID	Count of Transaction_ID	Customer Segment (Measure)
C014	10	High-Tier
C049	10	High-Tier
C099	10	High-Tier
C132	10	High-Tier
C160	10	High-Tier
C009	9	High-Tier
C013	9	High-Tier
C053	9	High-Tier
C072	9	High-Tier
C081	9	High-Tier
C129	9	High-Tier
C137	9	High-Tier
C149	9	High-Tier
C176	9	High-Tier
C180	9	High-Tier
C182	9	High-Tier
C196	9	High-Tier
C026	8	Mid-Tier
C047	8	Mid-Tier
C087	8	Mid-Tier
C166	8	Mid-Tier
C197	8	Mid-Tier
C001	7	Mid-Tier
C007	7	Mid-Tier
C015	7	Mid-Tier
C022	7	Mid-Tier
C023	7	Mid-Tier

```

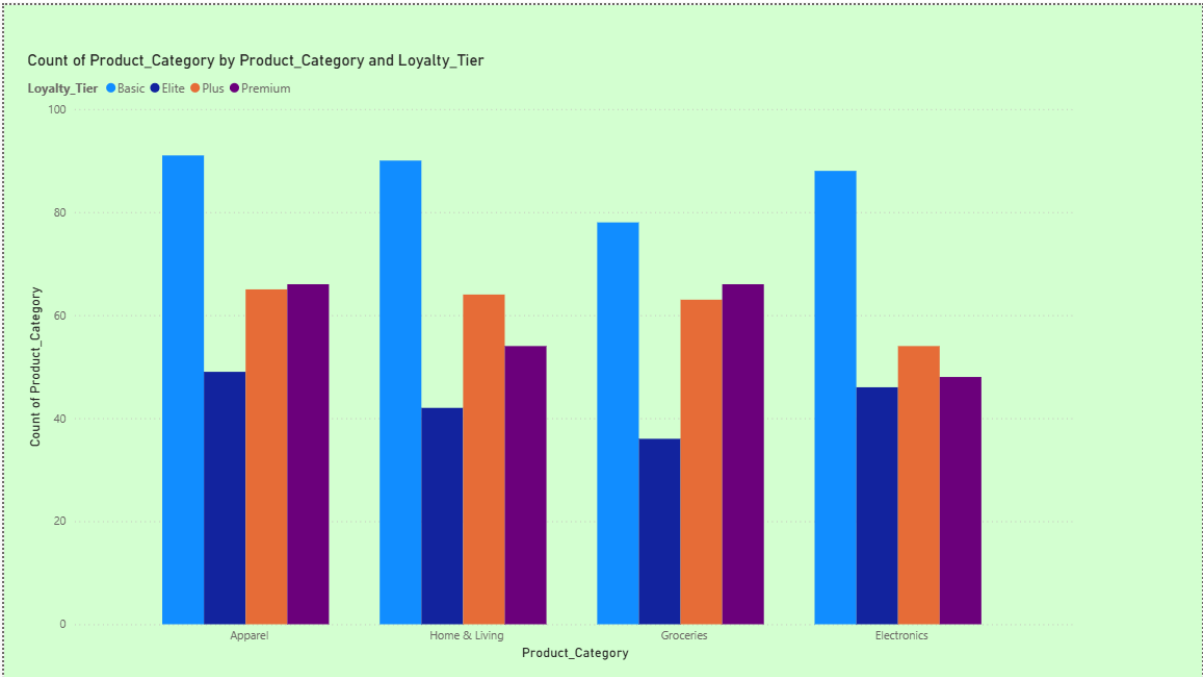
Customer Segment (Measure) =
VAR PurchaseCount =
    COUNT(Customer_Transactions[Transaction_ID])
RETURN
SWITCH(
    TRUE(),
    PurchaseCount <= 3, "Low-Tier",
    PurchaseCount <= 8, "Mid-Tier",
    "High-Tier"
)

```

- Compare avg. purchase frequency by Region, Age Group, Loyalty Tier

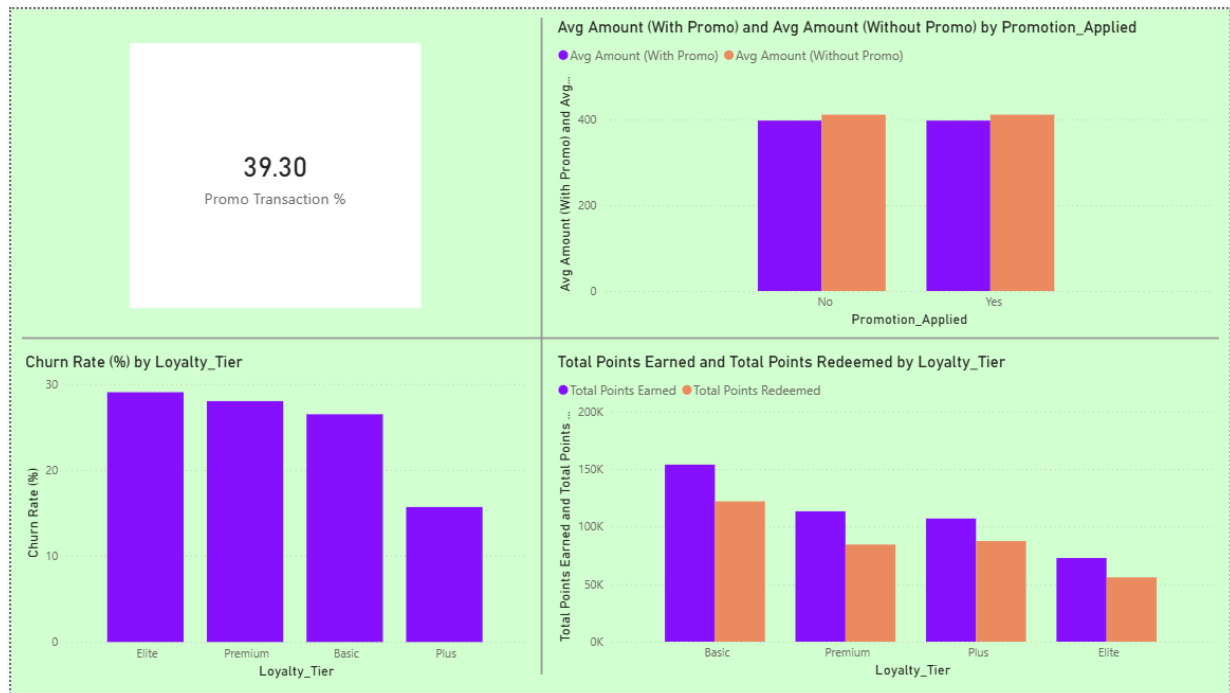


- Identify most purchased product categories by loyal customers



Task 4: Promotion & Loyalty Impact

- Calculate % of transactions with promotion applied.
- Compare average purchase amount with vs. without promotions.
- Analyze churn rate across loyalty tiers.
- Visualize Points Earned vs. Redeemed by Tier (Clustered Column Chart).



```
Promo Transaction % =
DIVIDE(
    [Promo Transactions],
    [Total Transactions],
    0
) * 100
```

```
Churn Rate (%) =
DIVIDE(
    [Churned Customers],
    [Total Customers],
    0
) * 100
```

```
Avg Amount (Without Promo) =  
CALCULATE(  
    [Avg Purchase Amount],  
    Customer_Transactions[Promotion_Applied] = "No"  
)
```

```
Avg Amount (With Promo) =  
CALCULATE(  
    [Avg Purchase Amount],  
    Customer_Transactions[Promotion_Applied] = "Yes"  
)
```

```
Total Points Earned =  
SUM(Loyalty_Program[Points_Earned])
```

```
Total Points Redeemed =  
SUM(Loyalty_Program[Points_Redeemed])
```

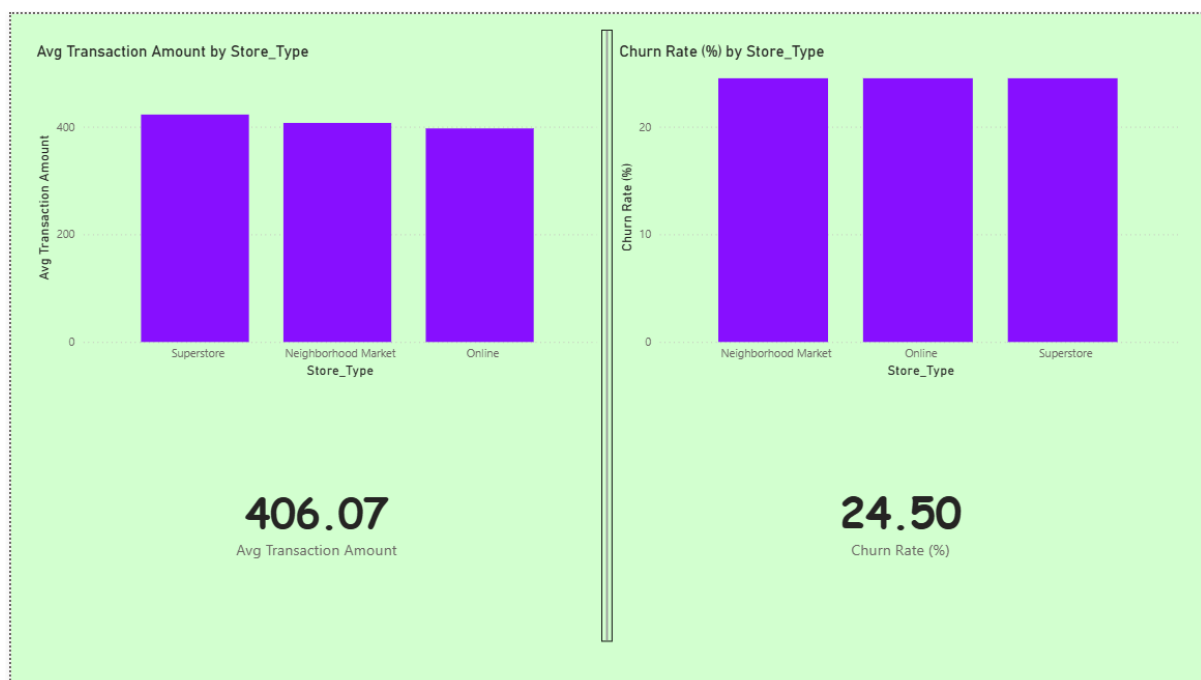
- Give recommendations to improve redemption & retention.
 - Simplify points redemption by lowering minimum limits, allowing partial redemption, and auto-applying points to reduce customer friction.
 - Increase loyalty value through tier-exclusive rewards, early sale access, and experiential benefits.
 - Improve point usage with expiry reminders and personalized nudges.
 - Reduce churn using segment-based retention strategies and personalized offers driven by customer data.
 - Optimize promotions by focusing on high-retention categories and testing non-discount incentives.
 - Strengthen the loyalty program by fast-tracking tier upgrades, rewarding milestones, and gamifying progress to encourage repeat purchases.

Task 5: Store & Channel Performance vs Retention

- Merge store data with transactions

Transaction_ID	Customer_ID	Store_ID	Store_Location.Store_ID	Product_Category	Transaction_Date	Year	Month Name	Amount
1 T0857	C001	S15	S15	Apparel	29-06-2024	2024	June	54
2 T0454	C001	S12	S12	Home & Living	16-03-2023	2023	March	5
3 T0818	C001	S14	S14	Electronics	23-10-2024	2024	October	63
4 T0550	C001	S02	S02	Electronics	02-07-2024	2024	July	76
5 T0531	C001	S20	S20	Apparel	19-10-2024	2024	October	33
6 T0597	C001	S09	S09	Groceries	11-06-2024	2024	June	55
7 T0274	C001	S19	S19	Home & Living	08-11-2023	2023	December	46
8 T0558	C002	S18	S18	Electronics	24-03-2024	2024	March	23
9 T0836	C002	S11	S11	Groceries	18-02-2024	2024	February	36
10 T0580	C002	S02	S02	Apparel	27-05-2023	2023	May	26
11 T0800	C002	S07	S07	Apparel	14-03-2023	2023	March	36
12 T0024	C002	S14	S14	Groceries	25-03-2024	2024	March	10
13 T0805	C003	S17	S17	Groceries	26-10-2023	2023	October	2
14 T0797	C003	S01	S01	Apparel	05-01-2024	2024	January	5
15 T0596	C004	S20	S20	Apparel	21-06-2023	2023	August	32
16 T0776	C004	S06	S06	Groceries	23-07-2024	2024	July	10
17 T0530	C005	S15	S15	Home & Living	21-10-2023	2023	October	67
18 T0758	C005	S09	S09	Electronics	14-08-2024	2024	August	12
19 T0355	C005	S13	S13	Apparel	06-08-2024	2024	August	21
20 T0544	C005	S16	S16	Groceries	29-05-2023	2023	May	60
21 T0023	C005	S09	S09	Groceries	11-12-2023	2023	December	14
22 T0631	C006	S15	S15	Home & Living	06-07-2024	2024	July	39
23 T0706	C006	S02	S02	Home & Living	21-05-2024	2024	May	8
24 T0243	C006	S01	S01	Electronics	06-06-2024	2024	June	30
25 T0202	C006	S01	S01	Apparel	07-08-2024	2024	August	9
26 T0651	C006	S03	S03	Apparel	16-05-2024	2024	May	3
27 T0035	C006	S07	S07	Apparel	21-06-2024	2024	June	5
28 T0883	C007	S10	S10	Apparel	08-08-2023	2023	August	32
29 T0210	C007	S05	S05	Groceries	11-05-2024	2024	May	36
30 T0669	C007	S10	S10	Apparel	16-02-2024	2024	February	5

- Visualize:



```
Avg Transaction Amount =
AVERAGE(Customer_Transactions[Amount])
```

```
Churn Rate (%) =
DIVIDE(
    [Churned Customers],
```

```

    [Total Customers],
    0
) * 100

```



```

Retention Rate (%) =
DIVIDE(
    [Retained Customers],
    [Total Customers (Store)],
    0
) * 100

```

Task 6: Customer Lifetime Value

- $CLV = \text{Total Amount Spent} / \text{Membership Duration (Years)}$

Membership_Duration_year	Customer_ID	CLV
7	C006	107.11
7	C008	120.65
7	C012	258.16
7	C014	463.16
7	C036	284.97
7	C041	113.96
7	C042	147.50
7	C047	429.08
7	C069	291.59
7	C071	70.29
7	C074	346.61
7	C077	426.53
7	C078	128.17
7	C081	403.44
7	C086	240.04
7	C097	343.47
7	C099	388.09
7	C105	210.58
7	C109	287.75
7	C115	366.42
7	C116	157.40
7	C123	111.45
7	C133	71.37
7	C135	438.96
7	C150	165.08
7	C151	253.45
7	C174	237.76
Total		6,863.03

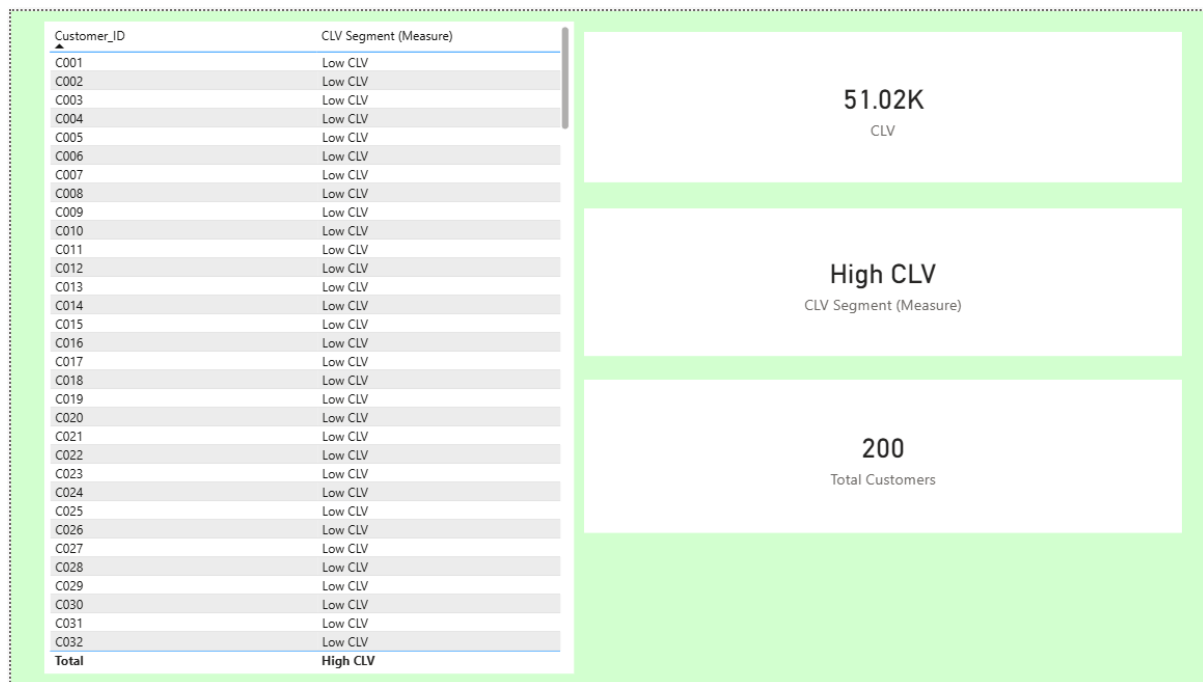
Membership_Duration_year
 7

6.86K
 CLV

```

CLV =
DIVIDE(
    [Total Amount Spent],
    [Membership Duration (Years)],
    0
)
  
```

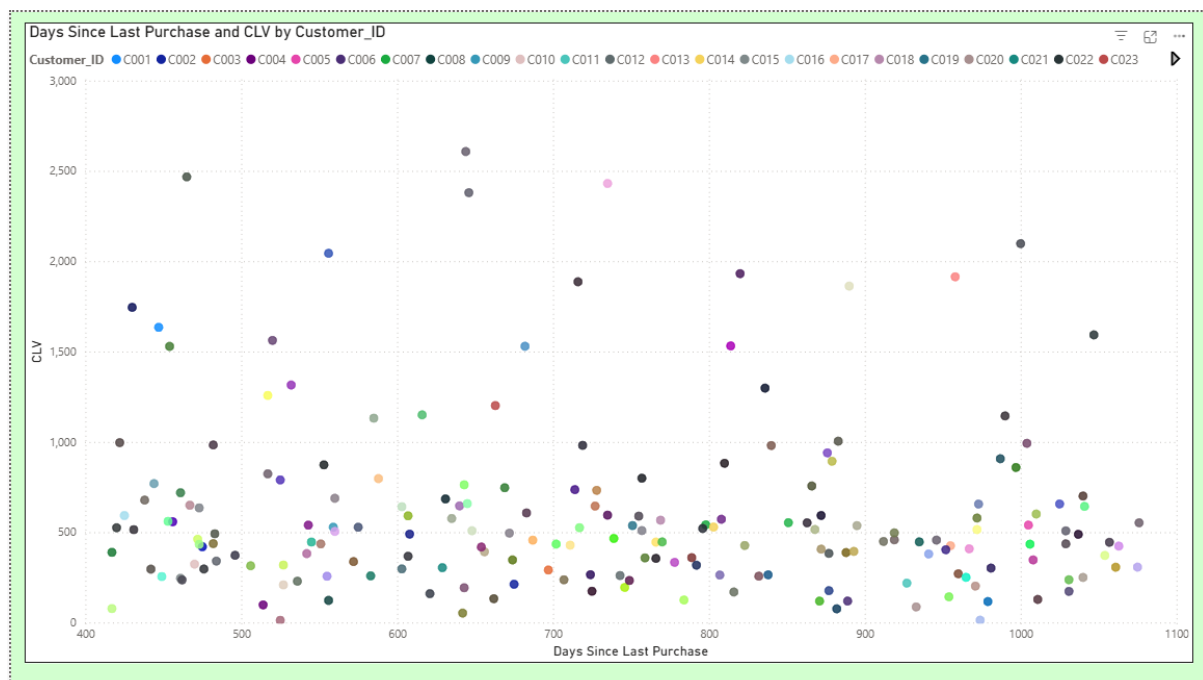
- Segment customers into Low, High CLV
 - Above Average CLV as High
 - Below Average CLV value- Low



● Visualize:

○ CLV vs Days Since Last Purchase

○ CLV by Loyalty Tier & Region

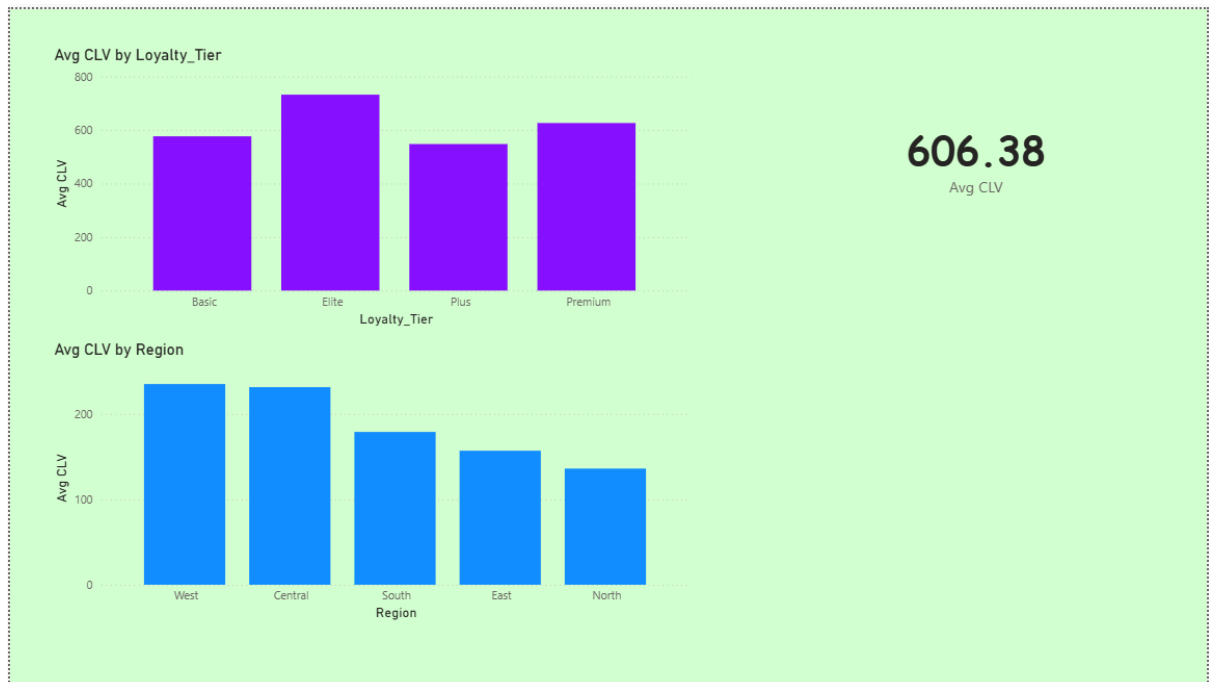


Days Since Last Purchase =

```

DATEDIFF(
    MAX(Churn_Labelled_Customers[Last_Purchase_Date]),
    TODAY(),
    DAY
)

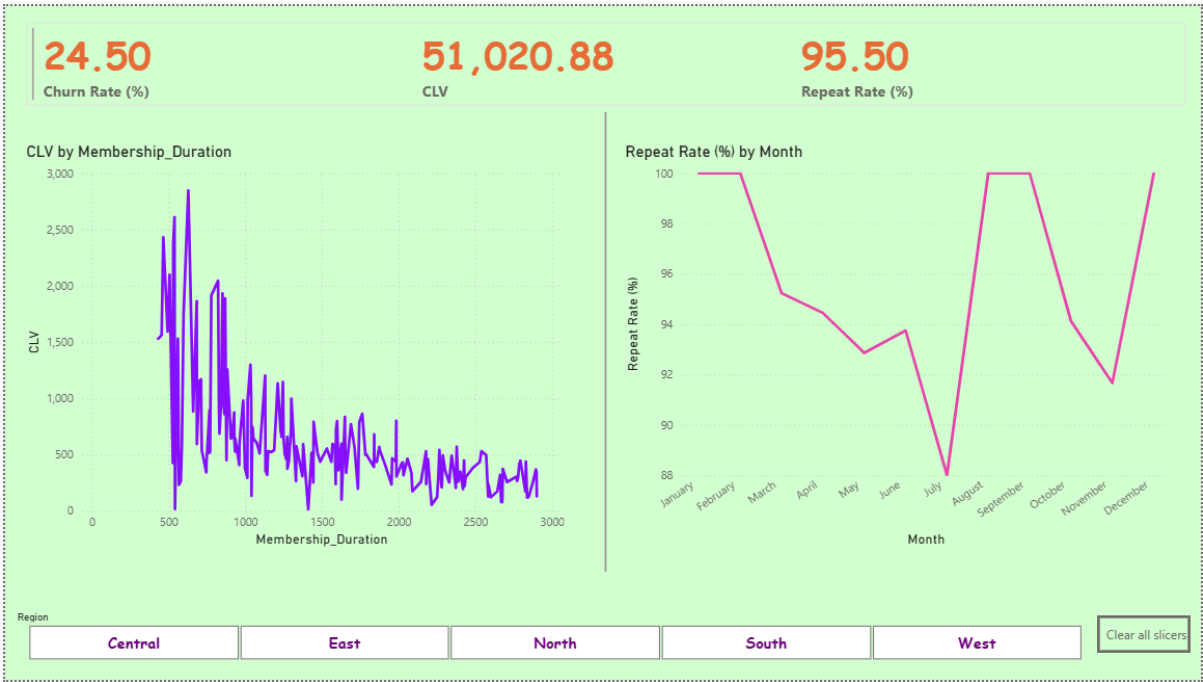
```



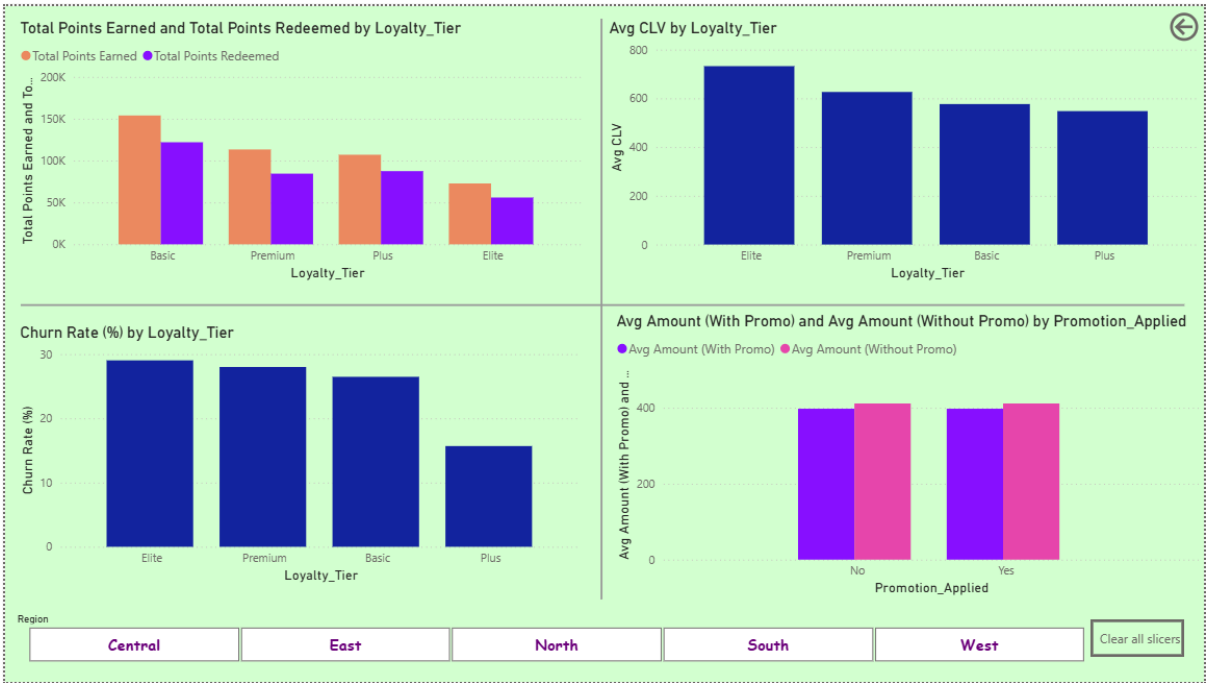
```
CLV =  
DIVIDE(  
    [Total Amount Spent],  
    [Membership Duration (Years)],  
    0  
)
```

Task 7: Final Dashboard & Executive Summary

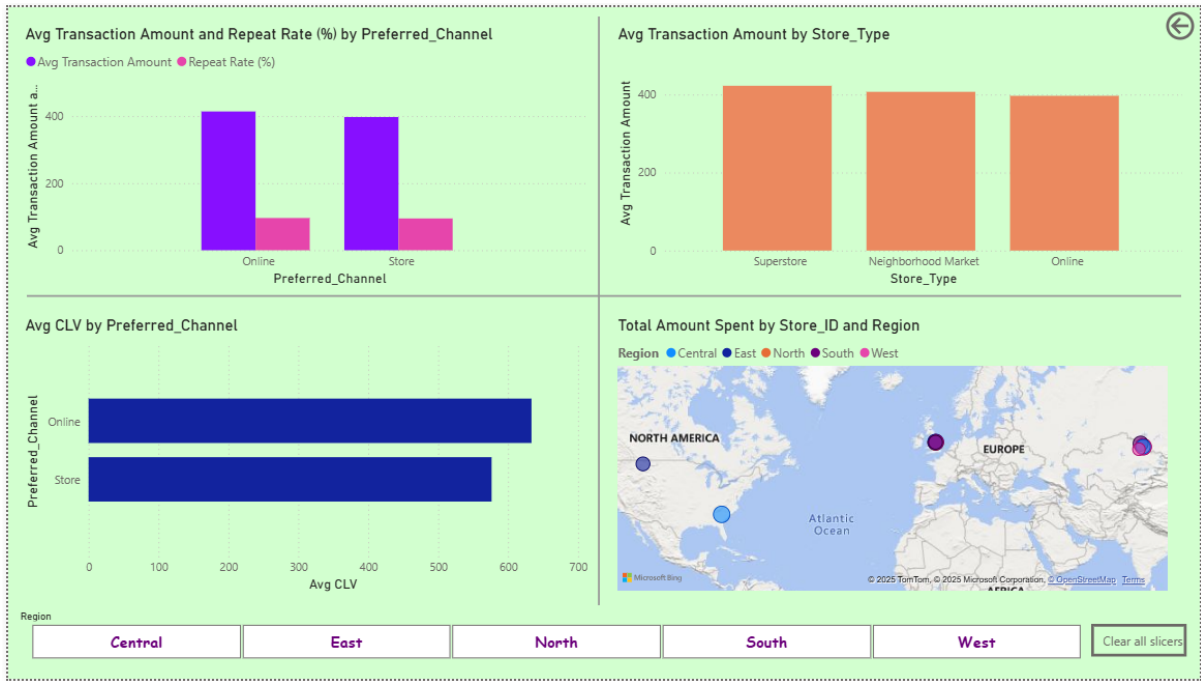
- Multi-page Power BI Report:
- Page 1: KPIs (Churn, CLV, Repeat Rate)



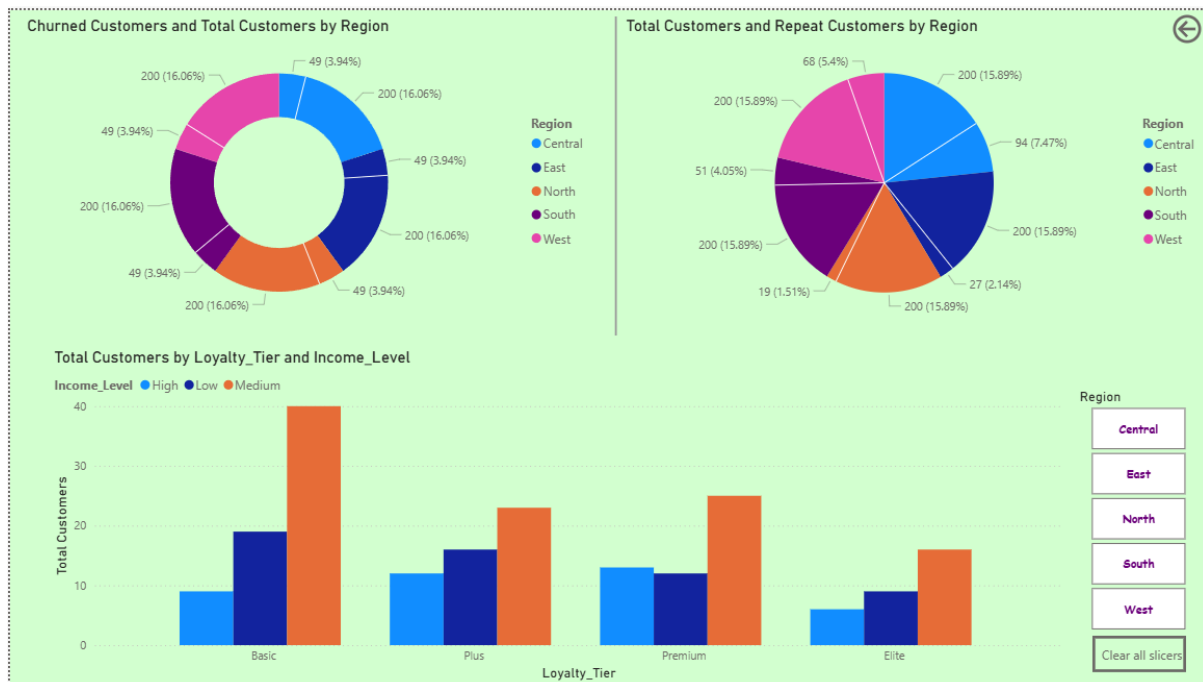
o Page 2: Loyalty & Promotion Impact



o Page 3: Store/Channel Insights



o Page 4: Segmentation (Churned, Repeat, High-Value)



- Slicers: Region, Channel, Income, Loyalty Tier

